

Lab 2: File System Operations and Asynchronous Programming

Overview

In this lab, you'll practice fundamental Node.js concepts including file operations, asynchronous programming, and building a simple HTTP server.

Prerequisites:

- Basic JavaScript knowledge
- VS Code installed
- Node.js installed

Learning Objectives

- Practice JavaScript fundamentals in a Node.js context
- Understand synchronous vs asynchronous file operations
- Build a simple HTTP server using Node.js
- Perform CRUD operations using file-based storage

Part 1: File Operations

Create a file named `fileOperations.js`. Use the provided `studentData` to create `students.json`.

Tasks

1. Create `students.json` with the sample data.
2. Implement both synchronous and asynchronous file operations.
3. Create functions to write, read, add a student, update a student's course (Bonus), and delete a student (Bonus).
4. Compare synchronous vs asynchronous operations (Bonus).
5. Use asynchronous file operations with a callback function (Bonus).

Requirements

- Implement both `readFileSync` and asynchronous `readFile`.
- Handle errors appropriately.
- Add comments explaining the differences between synchronous and asynchronous operations.

Part 2: HTTP Server

Build a simple HTTP server using Node.js's built-in `http` module. Use `students.json` as a file-based database.

Endpoints

Method	Endpoint	Description
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GET	/	Welcome message
GET	/students	Return all students
GET	/students/:id	Return student by ID
POST	/students	Add student
PUT	/students/:id	Update student
DELETE	/students/:id	Delete student

Server Requirements

- Use only built-in modules (http, fs, url).
- Use asynchronous readFile/writeFile for CRUD operations.
- Store data in students.json.
- Return JSON responses.
- Handle invalid routes with 404 Not Found.
- Handle invalid request bodies with 400 Bad Request.
- Return 201 Created when adding a student.
- Return 404 if a student does not exist.
- Use appropriate HTTP status codes.

Bonus

- Create student's
- update student

Run the Server

node server.js

Server URL:

http://localhost:3000